

EEG Sleep Stage Classification – Phase 2

Project Overview

This repository contains the **Phase 2 data pipeline** for an EEG Sleep Stage Classification project.

The goal of this phase is to build a **clean and reproducible data pipeline** that prepares EEG features for machine learning.

Phase 2 focuses on:

- Data preprocessing
- Feature engineering
- Creating a model-ready dataset

Important scope notes:

- Model training is **NOT** part of Phase 2
- Real-time systems are not implemented in this phase
- All

file paths must remain **relative** (no local absolute paths)

Repository link:

[[GITHUB REPOSITORY LINK HERE](#)]

Roles in Phase 2

This phase is organized around three roles.

Core Lead

The Core Lead builds the **data processing pipeline**.

Responsibilities:

- Implement preprocessing for EEG feature data
- Implement feature engineering
- Ensure data quality and label consistency
- Produce the final datasets used in later phases

Main outputs created by the Core Lead:

- outputs/preprocessed_features.csv
- outputs/model_ready_dataset.csv

Database Lead

The Database Lead works with the SQL database and analysis queries.

Responsibilities:

- Maintain the database file
- Write SQL queries to analyze sleep stage data
- Document schema and queries

Typical tasks include:

- Stage distribution analysis
- Band power statistics by sleep stage
- Row counts per subject

DevOps

The DevOps role ensures the pipeline runs reliably.

Responsibilities:

- Containerization using Docker
- CI pipeline setup

- Running the pipeline automatically in GitHub Actions

Environment Setup

To avoid system Python restrictions, the project should run inside a virtual environment.

Create a virtual environment:

Linux / macOS / WSL:

```
python3 -m venv .venv
source .venv/bin/activate
```

Windows (PowerShell):

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
```

Install required libraries:

```
python -m pip install pandas numpy
```

These packages can also be added to requirements.txt for reproducibility.

Input Data

The main dataset used in Phase 2 is:

```
data/sample/sleep_edf_sample_features_subject0.csv
```

This dataset already contains EEG features extracted from signals.

Typical columns include:

- subject_id
- epoch_id

- start_time_sec
- eeg_channel
- mean
- std
- min
- max
- signal_energy
- delta_power
- theta_power
- alpha_power
- beta_power
- sleep_stage
- sleep_stage_raw

Important columns for feature engineering:

- delta_power
- theta_power
- alpha_power
- beta_power
- signal_energy
- sleep_stage

Data Pipeline Overview

The Phase 2 pipeline consists of two main scripts.

Step 1 – Preprocessing

Step 2 – Feature Engineering

Data flow:

Input dataset

→ preprocessing

→ intermediate dataset

→ feature engineering

→ final datasets

Step 1: Preprocessing

Script:

```
scripts/preprocess.py
```

Purpose:

- Load the sample EEG feature dataset
- Perform basic cleaning
- Validate sleep stage labels

Tasks performed:

- Remove rows with invalid sleep stage labels
- Handle missing values when necessary
- Keep all important EEG feature columns

Input:

```
data/sample/sleep_edf_sample_features_subject0.csv
```

Output:

```
outputs/preprocessed_intermediate.csv
```

Run the preprocessing step:

```
python scripts/preprocess.py
```

After running successfully, the cleaned dataset is saved in the outputs directory.

Step 2: Feature Engineering

Script:

```
scripts/feature_engineering.py
```

Purpose:

Create additional EEG features that improve model performance.

Input dataset:

```
outputs/preprocessed_intermediate.csv
```

Outputs generated:

- `outputs/preprocessed_features.csv`
- `outputs/model_ready_dataset.csv`

Engineered Features

The script calculates several derived features.

Total power

$$\text{Total Power} = \delta_{\text{power}} + \theta_{\text{power}} + \alpha_{\text{power}} + \beta_{\text{power}} + \rho$$

Relative band powers

- `relative_delta_power`
- `relative_theta_power`
- `relative_alpha_power`
- `relative_beta_power`

These represent the proportion of each band relative to total power.

Ratios

- `delta_theta_ratio`
- `alpha_beta_ratio`

Log feature

- `log_signal_energy`

A small epsilon value is used in calculations to prevent division by zero.

Label Encoding

Sleep stage labels are converted from text to integers.

Mapping:

- Wake $\rightarrow$ 0
- N1 $\rightarrow$ 1
- N2 $\rightarrow$ 2
- N3 $\rightarrow$ 3
- REM $\rightarrow$ 4

A new column is created:

- `sleep_stage_encoded`
-

Model-Ready Dataset

The final dataset used for machine learning is:

`outputs/model_ready_dataset.csv`

This dataset contains:

- Only numeric feature columns
- One target column (`sleep_stage_encoded`)

Columns such as IDs or text labels are removed, including:

- `subject_id`
- `epoch_id`
- `start_time_sec`
- `eeg_channel`
- `sleep_stage`
- `sleep_stage_raw`

The result is a clean dataset suitable for training models in Phase 3.

Quick Start

Run the entire pipeline with the following commands.

Step 1 – create environment:

```
python -m venv .venv
```

Step 2 – activate environment:

```
source .venv/bin/activate
```

Step 3 – install dependencies:

```
python -m pip install pandas numpy
```

Step 4 – run preprocessing:

```
python scripts/preprocess.py
```

Step 5 – run feature engineering:

```
python scripts/feature_engineering.py
```

After execution, the outputs folder will contain:

- `preprocessed_intermediate.csv`
- `preprocessed_features.csv`
- `model_ready_dataset.csv`

The file **model_ready_dataset.csv** is the final dataset produced in Phase 2.

Next Phase

In Phase 3, this dataset will be used for:

- Machine learning model training
- Model evaluation
- Performance comparison

Phase 2 ends with a clean and validated dataset ready for those steps.